SAMPLE

Earth's Sky: The Deep Sky South

Southern Hemisphere Deep Sky Objects

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Introduction

Half the universe is on sale every clear night, and the southern hemisphere holds the better stock. Stand anywhere south of the tropics on a dark night, look up, and you are facing the center of the Milky Way instead of its outskirts, watching two satellite galaxies drift over the pole instead of zero, and picking from the brightest globular star clusters, the grandest nebulae, and the nearest stars as a northerner simply cannot. The deep sky south is the sum of everything that lies below the equator: the star clouds of Sagittarius and Scorpius riding high, the huge southern constellations that carry the sky's strangest and loveliest objects, the Magellanic Clouds hanging like smoke, and the dark river of the Coalsack cutting through the Cross. For observers born under it, the southern sky is simply the sky. For the rest of the world it is the half of the cosmos their horizon has always hidden, and the reason generations of astronomers have packed instruments onto ships.

The south is not just a mirror image of the north with the seasons flipped. Its deep sky is genuinely richer. The Milky Way's bright central bulge, and the great star-forming complex around it, lie almost entirely in southern declinations, which is why the galaxy's most spectacular regions, from the Carina Nebula to the Lagoon, are southern or at least southern-biased. The two Clouds of Magellan, the nearest substantial external galaxies to our own, never rise for most of the northern hemisphere. The two brightest globular clusters in the sky, Omega Centauri and 47 Tucanae, and the four of the ten brightest stars, are southern possessions. When astronomers in the 1840s finally brought serious telescopes below the equator, they did not merely fill in a blank half of a catalog; they discovered that the southern sky had been holding back some of the most important objects in astronomy, objects that would end up measuring the scale of the universe and exploding in it.

This book is a tour of that inheritance. It follows the deep sky south from the first star catalogs of Lacaille and Dunlop to the radio arrays now spread across the deserts of Western Australia and the Karoo, from the story of how the southern constellations got their odd names to the reasons Chile's Atacama Desert has become the most valuable astronomical real estate on Earth. It is written for the reader under any sky: the southerner who wants to know what the bright patches overhead really are, and the northerner who wants to know what they are missing, and why every few decades the northern world ships another enormous telescope south to catch up.

The chapters run in the same pattern as the rest of this series. First the history: who first charted the deep southern sky, and why the southern constellations look so different from the northern ones. Then the science: what the objects of the far south have taught us, from the period-luminosity law that first measured the universe to the supernova that let us see inside a dying star. Then the practice: where on Earth to observe the southern deep sky, what the great observatories are doing with it, and what a traveler with binoculars should look for first. And finally the state of it all: the mysteries still open, the facilities being built now, and some of the strange trivia a sky this crowded has collected along the way.

A note on the phrase itself. The deep sky, in the observing sense, means everything beyond the solar system: star clusters, nebulae, and galaxies, the objects that reward a telescope rather than a glance. The south in the title is the hemisphere, and the pairing of the two is not symmetric. The great observing lists assembled in the eighteenth and nineteenth centuries were compiled by European astronomers working under European skies, and their deep-sky catalogs ran thin below the equator; the southern objects that made those catalogs, and the far greater number that never did, are the subject here. This book treats the deep sky south as a single territory, with its own history, its own science, and its own future, because in practice that is how astronomers have always treated it: as a distinct realm, entered by voyage.

Chapter 1: The Discovery of The Deep Sky South

The systematic discovery of the southern deep sky is a story about the bottom of the world being brought into the written record, one impossible expedition at a time. For most of recorded history, astronomy's center of gravity lay in the northern hemisphere, and everything below the equator was rumor. Ptolemy's star catalog, compiled in the second century at Alexandria, reached down to the stars of Centaurus and Argo that grazed his southern horizon and stopped there; everything beyond was literally beneath his world. The southern constellations that Ptolemy's sky could not see were eventually named not by ancient mythmakers but by a French abbe with a small telescope and a large work ethic.

Nicolas-Louis de Lacaille arrived at the Cape of Good Hope in 1751, sent by the Academy of Sciences in Paris to measure the positions of southern stars and, ideally, to test methods for the longitude problem by observing Mars.

He set up on a site near Table Mountain, worked with refractors that would embarrass a modern amateur, and in two years of punishing routine he measured the positions of nearly ten thousand stars and catalogued forty-two nebulae and star clusters that no one had ever recorded, publishing the list in 1755. He also filled the sky's southern emptiness with constellations named for the instruments of the scientific age: the Telescope, the Microscope, the Air Pump, the Octant, the Clock, the Painter's Easel. It was less whimsy than necessity and more necessity than humility. The southern sky had almost no inherited mythology because almost no one who could write had seen it, and so the Enlightenment simply labeled it with its own tools. The names stuck, which is why a chart of the far south reads today like an inventory of an eighteenth-century laboratory.

The next great advance came from Australia. James Dunlop, a Scottish-born astronomer working at the observatory at Parramatta near Sydney, swept the southern sky from 1823 onward and in 1826 published a catalogue of 629 nebulae and clusters. It was a feat done with a homemade nine-inch reflector and, by later assessment, an eye that averaged double stars into single nebulae more often than anyone would now like; a good share of his entries correspond to nothing identifiable. But buried in the noise were many of the true showpieces, objects that carry his designation to this day, including the bright globular cluster now known as NGC 4833 and one of the sky's loveliest planetary nebulae. The catalogue was proof that the southern deep sky could reward a determined observer with equipment that a modern bystander would step over.

Then came the definitive survey. John Herschel, son of the man who had swept the northern deep sky into existence, decided that his father's work would be incomplete until the southern hemisphere had been treated with the same care. Shipping his own twenty-foot reflector to the Cape, he spent the years 1834 to 1838 at Feldhausen on the slopes of Table Mountain, cataloguing more than seventeen hundred nebulae and clusters and tens of thousands of stars, dissecting the Magellanic Clouds and the eta Carinae region, and confirming that the far south was not a lesser copy of the north but a region of the universe with its own treasures. His Results of Astronomical Observations, published in 1847, did for the southern sky what his father's catalogs had done for the northern: it made the objects a permanent part of science.

The photographic century that followed made the south central rather than peripheral, because the southern sky holds the standards that calibrate all of astronomy. Harvard College Observatory established a southern station at Arequipa, Peru, in the 1890s, partly so that the Magellanic Clouds could be photographed routinely, and those plates are where Henrietta Swan Leavitt found her variable stars. Her period-luminosity law, first published in 1908 and extended in 1912, grew out of southern hemisphere plates, and it became the yardstick with which the universe itself would be measured. Later, Harvard's Boyden Station moved to South Africa, the Lick Observatory ran a southern station in Chile, and the Royal Observatory at the Cape kept the longest continuous watch on southern skies on Earth. By the early twentieth century the center of gravity had already begun to tilt: the universe's biggest questions now pointed south.

The twentieth century institutionalized the migration. The European Southern Observatory was founded in 1962 by countries that owned no southern territory of their own, precisely because the sky's best real estate was in Chile; its first mountain, La Silla, saw first light in the late 1960s. Australia built the 3.9-meter Anglo-Australian Telescope at Siding Spring, opened in 1974. South Africa raised the Southern African Large Telescope at Sutherland in the 2000s, then the largest optical telescope on the continent. The discovery of the deep sky south began as a cataloguing project for whoever could endure the voyage, and it has ended with a ring of some of the largest machines humans have ever built, all pointed at the half of the sky that the ancients never named.

The nineteenth century also gave the southern sky its photographic patrimony. When the international Carte du Ciel project divided the sky among the world's observatories at the end of the century, the southern zones fell to the Cape, Sydney, Melbourne, Perth, and La Plata, and the southern half of the resulting Astrographic Catalogue was assembled as patiently as any northern zone. The twentieth century carried the photographic tradition further with the great southern sky atlases of the 1970s and 1980s, produced jointly by European and Australian observatories, which put the entire southern sky on photographic plates at uniform depth for the first time. Those atlases, scanned and digitized decades later, remain in use as historical baselines: a century of southern sky change can be measured against them.

Before Lacaille, the Dutch had already opened the southern sky to European science. On the first Dutch expedition toward the East Indies in the 1590s, the navigators Pieter Dirkszoon Keyser and Frederick de Houtman measured the positions of southern stars from the deck of their ship, and from their measurements the twelve southern constellations entered the record: the Phoenix, the Toucan, the Crane, the Peacock, the Bird of Paradise, the Fly, the Chamaeleon, the Flying Fish, the Water Snake, the Goldfish, the Southern Triangle, and the Indian. They

appeared first on the globes of the cartographer Plancius and then in Bayer's star atlas of 1603, exotic birds and fish flocking across a sky no European chart had ever reached. That is the other reason the southern constellations read so strangely: they are not an ancient mythology but a merchant marine's field guide, drawn from the decks of the first ships to stay south long enough to matter.